1 Integration of Differential Forms

1.1 Partition of Unity

Definition 1.1 (Support)**.**
Let X be a topological space, and $f : X \to \mathbb{R}$. Then the support of f is defined as

$$\operatorname{supp} f := \{\, x \in X \mid f(x) \neq 0 \,\}.$$

Theorem 1.1 (Partition of Unity)**.**
Let $\mathcal{M}$ be a C^∞ manifold with dimension m with atlas Φ.
Let $\Phi = \left\{\, \phi_j \;\middle|\; \phi_j : V_j \to \phi_j(V_j), j \in J \,\right\}$.
Then it is possible to construct a set of C^∞ functions $\rho_j, j \in J$ s.t.

$$1 = \sum_{j \in J} \rho_j, \ \operatorname{supp} \rho_j \subseteq V_j$$

1.2 Orientation

1.2.1 Definition

Definition 1.2 (Compatible Coordinate Charts)**.**
Given a manifold $\mathcal{M}$, its two coordinate charts are called compatible (have the same orientation) if,

$$\det J > 0.$$

Where J is the Jacobian matrix **??**.
If the manifold has a maximal compatible atlas, then we say the manifold is orientable, and we may call its corresponding orientation positive and denote the atlas Φ_+.

Theorem 1.2.
A manifold has either no orientation (any atlas is not compatible) or two orientations.

Theorem 1.3 (Orientability and Existence of Forms of Highest Degree).
A manifold is orientable iff there exists a nowhere vanishing differential form of the highest degree.

1.2.2 Positively Oriented Boundary

Definition 1.3 (Positively Oriented Boundary).
Let $\mathcal{M}$ be a orientable C^∞ manifold with dimension m, positively oriented by compatible atlas Φ_+. Define coordinate charts on $\partial\mathcal{M}$ from Φ as follows,

$$\phi^{\partial\mathcal{M}} : U_\phi \cap \partial\mathcal{M} \to \mathbb{R}^{m-1},$$

Then $\Phi_+^{\partial\mathcal{M}} := \left\{ \phi^{\partial\mathcal{M}} \mid \phi \in \Phi_+ \right\}$ determines an orientation on $\partial\mathcal{M}$, called the positive orientation.

1.3 Pseudoforms (Densities)

1.3.1 Pseudoscalars

Definition 1.4 (Space of Pseudoscalars).
Given a manifold $\mathcal{M}$ and a point p on it. Choose an arbitrary orientation $o_p = \Phi_+$ of $T_p\mathcal{M}$. On the set $\mathbb{R} \times \{o_p, -o_p\}$, define an equivalence relation,

$$\equiv : (c, o_p) \equiv (-c, -o_p).$$

We denote $\tilde{P}_p := \mathbb{R} \times \{o_p, -o_p\}/\equiv$, and call it the space of pseudoscalars. Under a choice of local orientation o, we often denote its element $(c, o) \equiv (-c, -o)$ by $\tilde{c}_o$.

Theorem 1.4 (Pseudoscalar Vector Space)**.**
The space $\tilde{P}_p$ forms a vector space under the operations,

$$\begin{aligned}(a, o) + (b, o) &= (a + b, o),\\ c(a, o) &= (ca, o).\end{aligned}$$

Specifically, $\dim \tilde{P}_p = 1$, since $\forall \tilde{c}_o \in \tilde{P}_p$, $\exists\, c \in \mathbb{R}$ s.t. $\tilde{c}_o = c\tilde{1}_o$.

Theorem 1.5 (Pseudoscalar Vector Bundle)**.**
On the space $\tilde{P} = \bigcup_{p\in\mathcal{M}} \tilde{P}_p$, define the canonical projection $\pi : \tilde{P} \to \mathcal{M}$ by $\tilde{P}_p \mapsto p$.
For any chart $\phi : U \to \mathbb{R}^m \in \Phi$ of the underlying manifold $\mathcal{M}$, it induces an orientation on $T_p\mathcal{M}$, denoted by o_ϕ. We define the local trivialization by

$$\begin{aligned}\tilde{\phi} : \tilde{\pi}(U) &\to U \times \mathbb{R}\\ (p, (c, o_\phi)) &\mapsto (p, c).\end{aligned}$$

And this makes pseudoscalars a vector bundle on any manifold $\mathcal{M}$.

Theorem 1.6 (Pseudoscalars Under Coordinate Transformation)**.**
Given two charts $\phi : U \to \mathbb{R}^m, \psi : V \to \mathbb{R}^m \in \Phi$ of a manifold $\mathcal{M}$ and a point $p \in U \cap V$, the following relationship holds on the pseudoscalar space $\tilde{P}_p$ Definition 1.4,

$$\tilde{c}_\phi = (\operatorname{sgn}\det J)\tilde{c}_\psi,$$

where $\tilde{c}_\phi$ means using the orientation induced by ϕ, and $\operatorname{sgn}\det J$ is the sign of the Jacobian matrix **??**.

Proof. If $o_\phi = o_\psi$, that is, $\det J > 0$ and their orientations agree, then $(c, o_\phi) \equiv (c, o_\psi)$. But if $o_\phi = -o_\psi$, that is, $\det J < 0$ and their orientations disagree, then $(c, o_\phi) \equiv (-c, o_\psi)$.

This shows that pseudoscalars ”flip sign” under charts of different orientation. ▨

1.3.2 Definition

Definition 1.5 (Pseudoforms).
The space of pseudoforms $\tilde{\Lambda}^k(\mathcal{M})$ is defined by the space of differential forms tensored with (twisted by) the pseudoscalar bundle,

$$\tilde{\Lambda}^k(\mathcal{M}) = \bigwedge^k T^*\mathcal{M} \otimes \tilde{P}(\mathcal{M}).$$

This means, pseudoforms are defined locally as

$$\tilde{\omega} = \tilde{1}_\phi \omega_{i_1\dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

1.3.3 Pseudoforms and Forms on Orientable Manifolds

Theorem 1.7 (Conversion of Pseudoforms and Forms).
For a pseudoform $\tilde{\omega} \in \tilde{\Lambda}^k(\mathcal{M})$ on an orientable manifold $\mathcal{M}$ of degree m positively oriented by the atlas Φ_+, the positive orientation is chosen in a continuous manner. Then $\tilde{P}(\mathcal{M})$ is isomorphic to $\mathbb{R}$ by the positive orientation in a continuous manner, so we can say

$$\begin{aligned} \iota : \tilde{\Lambda}^k(\mathcal{M}) &\to \Lambda^k(\mathcal{M}) \\ \tilde{1}_\phi \omega_{i_1\dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} &\mapsto o(\phi)\omega_{i_1\dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \end{aligned}$$

defines a smooth differential form on $\mathcal{M}$. Moreover, we can also convert it back,

$$\begin{aligned} \iota^{-1} : \Lambda^k(\mathcal{M}) &\to \tilde{\Lambda}^k(\mathcal{M}) \\ \omega_{i_1\dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} &\mapsto \tilde{1}_\phi o(\phi)\omega_{i_1\dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}. \end{aligned}$$

1.3.4 On Absolute Values

Remark.
Some people would say for $m = \dim \mathcal{M}$, pseudo-m-forms are absolute values of m-forms. This literal definition is doable in our formulation, and is given by the following theorem. ♍

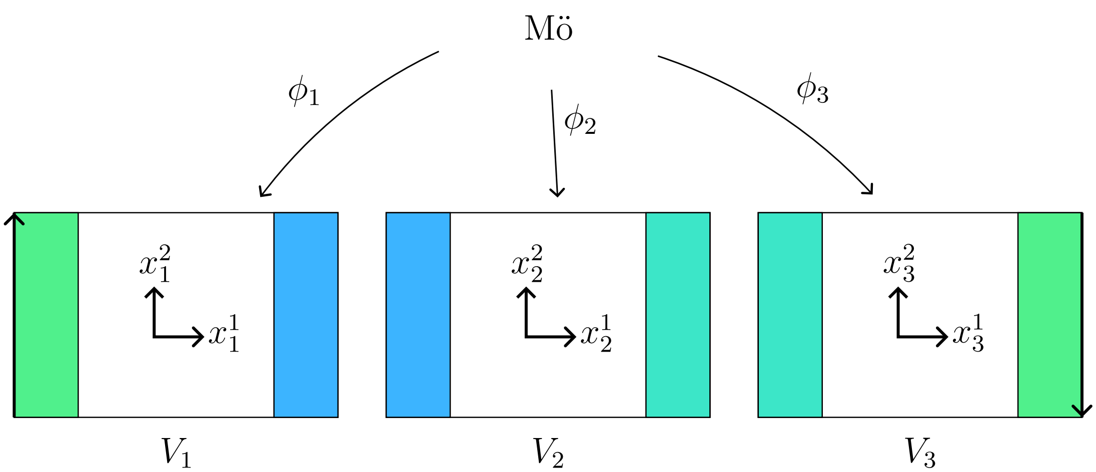
Theorem 1.8.

Given vectors $v_1, \dots, v_m \in T_p\mathcal{M}$, denote their orientation by o_v. Let $\tilde{o}_v(\tilde{1}_\phi) : \tilde{P}_p \to \mathbb{R}$ by $(c, o_v) \mapsto c$.

$$
\begin{aligned}
&\left(\tilde{o}_v(\tilde{1}_\phi) f(x) dx^1 \wedge \cdots \wedge dx^m\right)(v_1, \dots, v_m) \\
=&\left|\left(f(x) dx^1 \wedge \cdots \wedge dx^m\right)(v_1, \dots, v_m)\right|.
\end{aligned}
$$

Remark.

To understand these in a deeper level, we should think about why we cannot have a nowhere-vanishing differential form on a non-orientable manifold. Let's try to define a usual form on the Möbius band.



We create a Möbius band by patching three charts together, with overlapping areas indicated by the same color. $V_3 \cap V_1$ are patched together twisted, indicated by the reverse direction of arrows.

Let's say we have the three forms

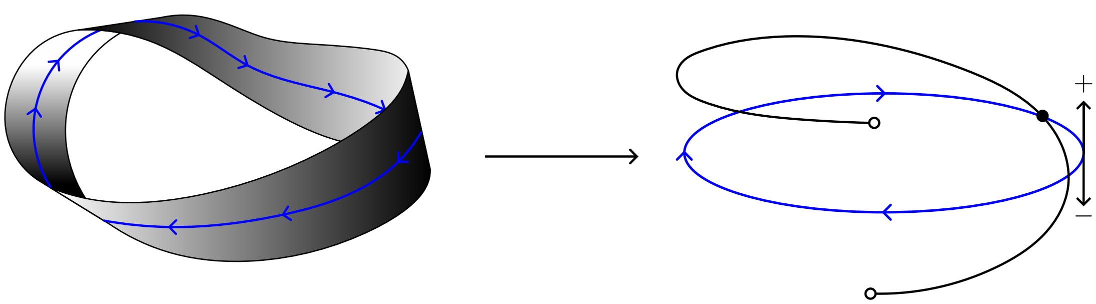
$$
\begin{aligned}
\omega_1 &= dx_1^1 \wedge dx_1^2 \\
\omega_2 &= f_2 dx_2^1 \wedge dx_2^2 \\
\omega_3 &= f_3 dx_3^1 \wedge dx_3^2
\end{aligned}
$$

defined on V_1, V_2, V_3, respectively, and with ω_1 prescribed to be 1 everywhere.

Then by the pullback criterion, **??**, we have

$$
\begin{aligned}
\omega_1 &= (\phi_2 \circ \phi_1^{-1})^* \omega_2 \\
&= f_2(\phi_2 \circ \phi_1^{-1}(x_1)) dx_1^1 \wedge dx_1^2 \\
\omega_2 &= (\phi_3 \circ \phi_2^{-1})^* \omega_3 \\
&= f_3(\phi_3 \circ \phi_2^{-1}(x_2)) dx_2^1 \wedge dx_2^2 \\
\omega_3 &= (\phi_1 \circ \phi_3^{-1})^* \omega_1 \\
&= -f_1(\phi_1 \circ \phi_3^{-1}(x_1)) dx_3^1 \wedge dx_3^2.
\end{aligned}
$$

But this cannot be, since by evaluating backwards from the last equation, we have $f_3 = -1$, $f_2 = f_3 = -1$, and $f_1 = f_2 = -1$. This contradicts with the given fact that $f_1 = 1$. By this reasoning, we have found an example of the fact that one cannot have a nowhere-vanishing usual form on a non-orientable manifold. Schematically, it looks like this,



where the black line resembles the value of ω. As you can see, it has a mismatch of sign, making it discontinuous. It is possible to resolve this situation by taking the absolute value, but if so, the linearity is broken. Introducing the pseudoscalar allows us to say $\tilde{1}_\phi = -\tilde{1}_\psi$ if their orientations are reversed, so that we can sort of "fold" the ends together, making it continuous again. In terms of coordinates,

$$
\begin{aligned}
\tilde{\omega}_3 &= (\phi_1 \circ \phi_3^{-1})^* \tilde{\omega}_1 \\
&= -\tilde{1}_{\phi_1} f_1(\phi_1 \circ \phi_3^{-1}(x_3)) dx_3^1 \wedge dx_3^2 \\
&= \tilde{1}_{\phi_3} f_1(\phi_1 \circ \phi_3^{-1}(x_3)) dx_3^1 \wedge dx_3^2 \\
&= \tilde{1}_{\phi_3} dx_3^1 \wedge dx_3^2.
\end{aligned}
$$

♍

1.3.5 Metric Volume Form

Definition 1.6 (The Metric Volume Form)**.**
On a smooth Riemannian manifold $\mathcal{M}$ of dimension m with metric g, the metric volume form is the unique pseudo-m-form that gives $\tilde{1}_\phi$ if the vectors are orthonormal and fits the local orientation of ϕ. Especially in 3-dimensional spaces, we denote it as vol.

Theorem 1.9 (Local Representation of Metric Volume Form)**.**
On a smooth Riemannian manifold $\mathcal{M}$ of dimension m with metric g and coordinate chart $\phi = x^1, \dots, x^m$, the metric volume form has local representation

$$\mathrm{vol} = \tilde{1}_\phi \sqrt{\det g}\; dx^1 \wedge \cdots \wedge dx^m,$$

where $\det g$ means $\det g_{ij}$, the determinant of the matrix of metric tensor.

Proof. Let $\mathrm{vol} = \tilde{1}_\phi dx^1 \wedge \cdots \wedge dx^m$ with the tangent vectors $(\partial_1)_p, \dots, (\partial_m)_p$ orthonormal in the sense of g. Then $g_{\mu\nu} = \delta_{\mu\nu}$ in that basis.

Under another coordinates $\psi = (y^1, \dots, y^m)$, we have

$$\begin{aligned}
\mathrm{vol} &= \tilde{1}_\phi \det\left(\frac{\partial x}{\partial y}\right) dy^1 \wedge \cdots \wedge dy^m \\
&= \tilde{1}_\psi \left|\det\left(\frac{\partial x}{\partial y}\right)\right| dy^1 \wedge \cdots \wedge dy^m.
\end{aligned}$$

But

$$\begin{aligned}
g &= \delta_{ij} dx^i \otimes dx^j \\
&= \delta_{ij} \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} dy^\alpha \otimes dy^\beta
\end{aligned}$$

by **??**. So

$$\begin{aligned}
g_{\alpha\beta} &= \sum_k \frac{\partial x^k}{\partial y^\alpha} \frac{\partial x^k}{\partial y^\beta} \\
&= \sum_k (J^{-1})_{k\alpha} (J^{-1})_{k\beta} \\
&= \sum_k ((J^{-1})^\top)_{\alpha k} (J^{-1})_{k\beta} \\
&= \left((J^{-1})^\top J^{-1}\right)_{\alpha\beta}.
\end{aligned}$$

Thus

$$\det g_{\alpha\beta} = \det\left((J^{-1})^\top J^{-1}\right) = \left(\det(J^{-1})\right)^2 = \left|\det \frac{\partial x}{\partial y}\right|^2$$

and the proof is completed. ▨

1.4 Integration of Forms of Highest Degree

Definition 1.7 (Integration)**.**
Let $\mathcal{M}$ be a paracompact C^∞ manifold of dimension m. Choose a C^∞ partition of unity $\rho_j, j \in J$ of $\mathcal{M}$ s.t. $\operatorname{supp} \rho_j \subseteq U_{\phi_j} := U_j$.
Let a pseudo-m-form $\tilde{\omega} \in \tilde{\Lambda}^m(\mathcal{M})$ has local expression $\tilde{\omega}_{\phi_j} = \tilde{1}_{\phi_j} f_j(x) dx_j^1 \wedge \cdots \wedge dx_j^m$.

$$\int_{\mathcal{M}} \tilde{\omega} = \sum_{j\in J} \int_{(-\infty,0]\times\mathbb{R}^{m-1}} (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$

if the finite sum exists and has the same value for all choices of ρ_j and ϕ_j.

Remark.
The following theorem reveals why we integrate pseudoforms, not usual forms.
♍

Theorem 1.10 (Criterion of Existence of Integral)**.**
If $\operatorname{supp} \tilde{\omega}$ is compact, then $\int_{\mathcal{M}} \tilde{\omega}$ exists.

Proof. Let two sets of coordinate charts be

$$\phi_j : U_j \to V_j, j \in J$$
$$\phi_k' : U_k' \to V_k', k \in K.$$

And cooresponding partition of unity be ρ_j, ρ_k'.

(The goal) Show

$$\sum_{j\in J} \int (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$
$$= \sum_{k\in K} \int (\rho_k' \circ {\phi_k'}^{-1})(x') f_k'(x') d{x'}^1 \dots d{x'}^m$$

(Split using ρ'_k)

$$\begin{aligned}\int_{\mathcal{M}} \tilde{\omega} &= \sum_{j\in J} \int (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \sum_{j\in J} \int \sum_{k\in K} (\rho'_k \circ \phi_j^{-1})(x)(\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m.\end{aligned}$$

Since the sum is finite, and $\operatorname{supp} \tilde{\omega}$ is compact, and therefore the integral is not improper; thus, there can be no limit or Fubini problems on exchanging sums and integrals. So

$$\int_{\mathcal{M}} \tilde{\omega} = \sum_{j\in J} \sum_{k\in K} \int (\rho_j \rho'_k \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$

(Change of variables) First fix j, k.

$$\begin{aligned}&\int (\rho_j \rho'_k \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ =& \int (\rho_j \rho'_k \circ \phi_j^{-1})(\phi_j \circ {\phi'_k}^{-1}(x')) f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \left|\det\left(\frac{\partial x}{\partial x'}\right)\right| d{x'}^1 \dots d{x'}^m \\ =& \int (\rho_j \rho'_k \circ {\phi'_k}^{-1}(x')) f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \left|\det\left(\frac{\partial x}{\partial x'}\right)\right| d{x'}^1 \dots d{x'}^m\end{aligned}$$

(Pullback Condition) From **??**,

$$\tilde{\omega}_{\phi'_k} = \left(\phi_j \circ {\phi'_k}^{-1}\right)^* \tilde{\omega}_{\phi_j}.$$

Using Theorem 1.6, we see,

$$\begin{aligned}&\tilde{1}_{\phi'_k} f'_k(x') d{x'}^1 \wedge \dots \wedge d{x'}^m \\ =&\tilde{1}_{\phi_j} f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \left(\frac{\partial x^1}{\partial {x'}^{\ell_1}} d{x'}^{\ell_1}\right) \wedge \dots \wedge \left(\frac{\partial x^m}{\partial {x'}^{\ell_m}} d{x'}^{\ell_m}\right) \\ =&\tilde{1}_{\phi_j} f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \sum_{\sigma\in S_m} (\operatorname{sgn}\sigma) \frac{\partial x^1}{\partial {x'}^{\sigma(1)}} \dots \frac{\partial x^m}{\partial {x'}^{\sigma(m)}} d{x'}^1 \wedge \dots \wedge d{x'}^m \\ =&(\operatorname{sgn}\det J) \tilde{1}_{\phi'_k} f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \det\left(\frac{\partial x}{\partial x'}\right) d{x'}^1 \wedge \dots \wedge d{x'}^m \\ =&\tilde{1}_{\phi'_k} f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \left|\det\left(\frac{\partial x}{\partial x'}\right)\right| d{x'}^1 \wedge \dots \wedge d{x'}^m.\end{aligned}$$

So we see

$$f'_k(x') = f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \left|\det\left(\frac{\partial x}{\partial x'}\right)\right|.$$

Therefore, the integral

$$\begin{aligned} &\int (\rho_j \rho'_k \circ {\phi'_k}^{-1}(x')) f_j(\phi_j \circ {\phi'_k}^{-1}(x')) \left| \det\left(\frac{\partial x}{\partial x'}\right) \right| d{x'}^1 \dots d{x'}^m \\ = &\int (\rho_j \rho'_k \circ {\phi'_k}^{-1}(x')) f'_k(x') d{x'}^1 \dots d{x'}^m \end{aligned}$$

(Closing) By moving the sum wrt $j \in J$ into the integral and using the property of partition of unity, the proof is completed. ▨

1.5 Integration of Forms of Lower Degree

1.5.1 Pseudoorientation (Transverse Orientation)

Remark.
Because pseudoscalars depend on local orientations, we need an extra structure in order to pullback pseudoforms onto a submanifold. ♍

Definition 1.8 (Pseudoorientation)**.**
For $\mathcal{Z}$ a submanifold of $\mathcal{M}$, pseudoorientation is a map that for all $p \in \mathcal{Z}$, takes a local orientation of $\mathcal{Z}$ at p (an orientation of $T_p\mathcal{Z}$) to a local orientation of $\mathcal{M}$ at p (an orientation of $T_p\mathcal{M}$) **continuously in p, and takes opposite orientations to opposite orientations**.

Remark.
Requiring well-defined surface normal (with a metric, of course) is precisely this criterion for $(m-1)$ dimensional submanifolds, for we can form an orientation of $\mathcal{M}$ from an orientation of $\mathcal{Z}$ (which is essentially $(m-1)$ vectors on $T_p\mathcal{Z}$) and the normal.
This makes Mö integrable as a submanifold of Mö $\times \mathbb{R}$, since we can choose the surface normal to be positive everywhere. But Mö is not integrable as a submanifold of $\mathbb{R}^3$, since if we try to assign surface normal along the strip, when we get back to where we started, the direction would be reversed. ♍

Definition 1.9 (Pullback of Pseudoscalars)**.**
Given a manifold $\mathcal{M}$ and its submanifold $\mathcal{Z}$, and ψ be a chart of $\mathcal{M}$, ϕ a chart of $\mathcal{Z}$. At the point p, if the pseudoorientation i Definition 1.8 is chosen such

that the local orientations $o_\phi \mapsto o_\psi$, then we say $i^*\tilde{1}_\psi = \tilde{1}_\phi$. If $o_\phi \mapsto -o_\psi$, then $i^*\tilde{1}_\psi = -\tilde{1}_\phi$.

1.5.2 Definition

Definition 1.10 (Integration of Lower Degree Forms)**.**
Let $\mathcal{Z}$ be an pseudooriented C^∞ submanifold of dimension d of $\mathcal{M}$, $i : \mathcal{Z} \to \mathcal{M}$ be the immersion map.
Let $\omega \in \Lambda^d(\mathcal{M})$, we define

$$\int_{\mathcal{Z}} \omega := \int_{\mathcal{Z}} i^*\omega$$

if it exists.

1.5.3 Stoke's theorem

Theorem 1.11 (Stoke's Theorem)**.**
If $\mathcal{M}$ is an oriented C^∞ manifold of dimension m and $\omega \in \Lambda^{m-1}(\mathcal{M})$, then

$$\int_{\mathcal{M}} d\omega = \int_{\partial \mathcal{M}} \omega := \int_{\partial \mathcal{M}} i^* \omega,$$

where $i : \partial\mathcal{M} \to \mathcal{M}$ is just the immersion map, $i : p \mapsto p$.

Proof. (Partition Using Charts) Choose a C^∞ partition of unity Theorem 1.1 $\rho_j, j \in J$ s.t. $\operatorname{supp} \rho_j$ are compact and $\operatorname{supp} \rho_j \subseteq U_{\phi_j} := U_j$.

Now $\omega = \sum_{j \in J} \rho_j \omega$ is a finite sum. So it suffices to show that, if $\eta \in \Lambda^{m-1}(\mathcal{M})$, $\operatorname{supp} \eta$ compact and $\operatorname{supp} \eta \subseteq U_\phi$ then $\int_{\mathcal{M}} d\eta = \int_{\partial\mathcal{M}} \eta$, and apply $\eta = \rho_j \omega$ for all $j \in J$.

(The integral) Suppose the coordinates of ϕ is labeled $x^1, \dots, x^m$. Locally, let

$$\eta = \sum_{\ell=1}^{m} f_\ell dx^1 \wedge \cdots \cancel{dx^\ell} \cdots \wedge dx^m,$$

where $f_\ell : (-\infty, 0] \times \mathbb{R}^{m-1} \to \mathbb{R}$, and it is C^∞. Then, also locally,

$$\begin{aligned} d\eta &= \sum_{\ell=1}^{m} \frac{\partial f_\ell}{\partial x^\ell} dx^\ell \wedge dx^1 \wedge \cdots \cancel{dx^\ell} \cdots \wedge dx^m \\ &= \left(\sum_{\ell=1}^{m} (-1)^{\ell-1} \frac{\partial f_\ell}{\partial x^\ell} \right) dx^1 \wedge \cdots \wedge dx^m. \end{aligned}$$

Then by Definition 1.7,

$$\int_{\mathcal{M}} d\eta = \int_{(-\infty,0]\times\mathbb{R}^{m-1}} \left(\sum_{\ell=1}^{m} (-1)^{\ell-1} \frac{\partial f_\ell}{\partial x^\ell} \right) dx^1 \dots dx^m.$$

(To be precise, we need to choose another partition of unity ρ_j' to do integration. But we can just choose it to cover all of $\operatorname{supp} \eta$ and don't care all other parts, so that doesn't matter too much.)

Choose a rectangular region R s.t.

$$\operatorname{supp} \eta \subseteq [a_1, 0] \times \cdots \times [a_m, b_m]$$

and define

$$R_\ell := [a_1, 0] \times \dots \cancel{[a_\ell, b_\ell]} \dots \times [a_m, b_m].$$

($\ell = 2, \dots, m$) In this case, by Fubini and FTC,

$$\begin{aligned}
&\int_R \frac{\partial f_\ell}{\partial x^\ell} dx^1 \dots dx^m \\
=&\int_{R_\ell} \left(\int_{a_\ell}^{b_\ell} \frac{\partial f_\ell}{\partial x^\ell} dx^\ell \right) dx^1 \dots \cancel{dx^\ell} \dots dx^m \\
=&\int_{R_\ell} (\underbrace{f_\ell(x^1, \dots, b_\ell, \dots, x^m)}_{0} - \underbrace{f_\ell(x^1, \dots, a_\ell, \dots, x^m)}_{0}) dx^1 \dots \cancel{dx^\ell} \dots dx^m \\
=&0,
\end{aligned}$$

since $\operatorname{supp} \eta \subseteq R$, so on the boundary $f = 0$.

($\ell = 1$) Now the integral has only one term left.

$$\begin{aligned}
\int_{\mathcal{M}} d\eta &= \int_R \frac{\partial f_1}{\partial x^1} dx^1 \dots dx^m \\
&= \int_{R_1} \left(\int_{a_1}^{0} \frac{\partial f_1}{\partial x^1} dx^1 \right) dx^2 \dots dx^m \\
&= \int_{R_\ell} (f_1(0, x^2, \dots, x^m) - \underbrace{f_1(a_1, x^2, \dots, x^m)}_{0}) dx^2 \dots dx^m \\
&= \int_{\mathbb{R}^{m-1}} (f_1 \circ i) dx^2 \dots dx^m \\
&= \int_{\partial \mathcal{M}} i^* \eta.
\end{aligned}$$

▨

1.5.4 Integrate Without Partition of Unity

> **Remark.**
> The truth is, no one can (or wants) to write down a partition of unity. If one do write it down, integrating it is a big problem.
> Fortunately, if one decompose any manifold $\mathcal{M}$ into finitely many submanifolds $\mathcal{M}_i, i \in I$ which intersect only on their boundaries, then $\int_{\mathcal{M}} \tilde{\omega} = \sum_{i \in I} \int_{\mathcal{M}_i} f_i^* \tilde{\omega}$, where f_i is the injective immersion map $f_i : \mathcal{M}_i \to \mathcal{M}$. ♍